INDEX

INSTITUTE

VISION MISSION

INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.

DEPARTMENT

VISION

MISSION

DEPARTMENT
VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

PROGRAM EDUCATIONAL OBJECTIVES, PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES

Program Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge**: Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. **[K3]**
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. **[K4]**
3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. **[K5]**
4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **[K5]**
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. **[K3]**
6. **The Engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. **[K3]**
7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. **[K3]**
8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. **[K3]**
9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. **[K6]**
10. **Communication**: Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. **[K2]**
11. **Project management and finance**: Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. **[K6]**
12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. **[K1]**

ACADEMIC CALENDAR

Grams: "TECHNOLOGY"
Email: dapjntuk@gmail.com

Phone: 0884-2300991
Mobile: +9963993504

Directorate of Academic & Planning
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA
KAKINADA-533003, Andhra Pradesh, INDIA
(Established by AP Government Act No. 30 of 2008)

Lr. No. JNTUK/DAP/AC/B. Tech/III Year/2019-20 *Date: 30-05-2019*

Dr. A. Mallikarjuna Prasad
M.E, Ph.D.,
Director, Academic Planning

To
All the Principals of Affiliated Colleges,
JNTUK, Kakinada

ACADEMIC CALENDAR FOR B.TECH III YEAR (2017 BATCH)

I SEMESTER			
Description	**From**	**To**	**Weeks**
Commencement of Class Work	**10.06.2019**		
I Unit of Instructions	10.06.2019	03.08.2019	8W
I Mid Examinations	05.08.2019	10.08.2019	1W
II Unit of Instructions	12.08.2019	05.10.2019	8W
II Mid Examinations	07.10.2019	12.10.2019	1W
Preparation & Practicals	14.10.2019	19.10.2019	1W
End Examinations	21.10.2019	02.11.2019	2W
Commencement of II Semester Class Work	**18.11.2019**		
II SEMESTER			
I Unit of Instructions	**18.11.2019**	11.01.2020	8W
I Mid Examinations	13.01.2020	23.01.2020	1W
II Unit of Instructions	24.01.2020	21.03.2020	8W
II Mid Examinations	23.03.2020	28-03-2020	1W
Preparation	30.03.2020	04.04.2020	1W
End Examinations	06.04.2020	18.04.2020	2W
Commence of IV Year Class Work	**08.06.2020**		

A.M.prasad
Director Academic Planning

Copy to the Secretary to the Hon'ble Vice Chancellor, JNTUK.
Copy to PA to the Rector, JNTUK.
Copy to PA to the Registrar, JNTUK.
Copy to PA to the Director of Evaluation, JNTUK.

CO-CURRICULAR AND EXTRA CURRICULAR ACTIVITIES

05/06/2019 – World Environment Day	**15/09/2019 – Engineers Day**		**21/03/2019 – International Forest Day**
21/08/2019 to 24/08/2019– 1^{st} year B.Tech. Introduction program	**16/09/2019 – World Ozone Day**	**22/12/2019 – National Mathematics Day**	**22/03/2019 –World Water Day**
15/08/2018 – Independence Day	**September – Intramurals**	**26/01/2019 – Republic Day**	**23/03/2019 – World Meteorological Day**
05/09/2019 – Teachers Day	**11/11/2019 – National Education Day**	**08/03/2019 – International Women's Day**	**24/03/2019 – Earth Hour**
	03/12/2019 – Antipollution Day		**In the month of March – Association Days**

SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science &Engineering (Accredited by NBA)

CLASS CONSOLIDATED TIME TABLE

Class: 3rd B.Tech 1st Semester   w.e.f. 10-06-2019

Section – A   Class Coordinator :Mr. N V Muralikrishna Raja   Room B-202

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	DBMS	OS	UP	CD	LUNCH BREAK	UML LAB		
Tue	UML	APTITUDE		DBMS		CD	UP	OS
Wed	OS	ECS		CD		PE&HV	DBS	UP
Thu	UP	UML	DBMS	OS		OSLP LAB		
Fri	CD	DBMS LAB				UML	TT	
Sat	PE&HV	CD	UML	UP		OS	DBMS	LIB/SPO

Section – B   Class Coordinator: Mrs. P. Praneetha   Room No: B-203

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	UP	CD	DBMS	OS	LUNCH BREAK	APTITUDE		PE&H
Tue	DBMS	OS	UML	CD		UML LAB		
Wed	PE&HV	DBMS LAB				UP	UML	CD
Thu	UML	ECS		UP		DBMS	OS	LIB
Fri	CD	UP	OS	DBMS		OSLP LAB		
Sat	OS	DBMS	TT LAB			CD	UML	SPORTS

Section – C   Class Coordinator: Mrs. A. LakshmiLavanya   Room No: B-301

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	DBMS	OSLP LAB			LUNCH BREAK	CD	OS	UP
Tue	UP	CD	OS	DBMS		DBMS LAB		
Wed	UML	APTITUDE		PE&HV		UP	ECS	
Thu	OS	UML LAB				DBMS	UML	CD
Fri	PE&HV	OS	CD	UP		UML	DBMS	LIB/SPO
Sat	CD	OS	TT LAB			UML	UP	DBMS

Section – D   Class Coordinator: Mr. I. Anandbabu   Room No: B-302

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	PE&HV	ECS		DBMS	LUNCH BREAK	CD	OS	UP
Tue	CD	OSLP LAB				UML	PE&HV	DBMS
Wed	DBMS	UML	OS	UP		APTITUDE		CD
Thu	OS	CD	UP	DBMS		UML	TT LAB	
Fri	UML	UML LAB				OS	UP	LIB
Sat	UP	OS	DBMS	CD		DBMS LAB		

Staff Details:

S. No.	Course Code	Course Name	Section A	B	C	D
1.	**R1631051**	**CD**	Mr.G.SriramGanesh	Mrs.P.Praneetha	Mrs.P.Praneetha	Mr.G.SriramGanesh
2.	**R1631052**	**UP**	Mr.I.Anand Babu	Dr.D.JayaKumari	Mrs.A.L.Lavanya	Mr.I.Anand Babu
3.	**R1631053**	**OOAD UML**	Mr.K.LakshmiNarayana	Mr.G.Nataraj	Mrs.K.RajaniKumari	Mr.A.Rajesh
4.	**R1631054**	**DBMS**	Mr.VKHanumanT	Ms.I.Lavanya	Sd.SabirHussain	Mr.D.Ganesh
5.	**R1631055**	**OS**	Mrs.G.Loshma	Mrs.Y.DivyaVani	Mr.Y. Ravi Raju	Mrs.G.Loshma
6.	**R1631056**	**UM Lab**	Mr.K.LakshmiNarayana, Mr.N V Muralikrishna Raja	Mr.G.Nataraj, Mr.N V Muralikrishna Raja	Mrs.K.RajaniKumari, Mrs.Y.DivyaVani	Mr.A.Rajesh, Mrs.K.RajaniKumari,
7.	**R1631057**	**OS&LP Lab**	Mrs.G.Loshma& Mr.I.Anand Babu	Mrs.Y.DivyaVani& Dr.D.JayaKumari	Mr.Y. Ravi Raju& Mrs.A.L.Lavanya	Mrs.G.Loshma& Mr.I.Anand Babu
8.	**R1631058**	**DBMS Lab**	Mr.VKHanumanT, Mrs.N. Hiranmayee	Ms.I.Lavanya, Mrs.N. Hiranmayee	Sd.SabirHussain, Mrs.P.Praneetha	Mr.D.Ganesh, Mr.N V Muralikrishna Raja
9.	**R1631049**	**PE & HV**	Mr.N V Muralikrishna Raja	Mr.Sd.SabirHussain	Mr.B.Madhava Rao	Mrs.Y.DivyaVani
10.		**ECS**	Mrs.T.Sujani	Mrs.K.V.L.B.Devi	Mrs.K.V.L.B.Devi	Mr.B.suresh
11.		**Aptitude**	Mr.J.N.V.Somyajulu	Mr.J.N.V.Somyajulu	Mr.J.N.V.Somyajulu	Mr.J.N.V.Somyajulu
12.		**TT**	R. Leela Phani Kumar	A. Venkanna Babu	R. Leela Phani Kumar	A. Venkanna Babu

Head of the Department

COURSE STRUCTURE

IIIB. Tech –I Semester

III Year –I Semester

S. No.	Subjects	L	T	P	Credits
1	Compiler Design	4	--	--	3
2	Unix Programming	4	--	--	3
3	Object Oriented Analysis and Design using UML	4	--	--	3
4	Database Management Systems	4	--	--	3
5	Operating Systems	4	--	--	3
6	Unified Modeling Lab	--	--	3	2
7	Operating System &Linux Programming Lab	--	--	3	2
8	Database Management System Lab	--	--	3	2
MC	Professional Ethics &Human Values	--	3	--	--
	Total Credits				21

LESSON PLANS

Compiler Design

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: III/I **Section: A,B,C& D**
Name of the Course: Compiler Design **Course Code:R1631051/301**

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, Student will be able to:

301.1	Discuss about Compiler Phases and token specification by lexical analyzer.	**K2**
301.2	Demonstrate the role of syntax analysis and different types of parsers.	**K3**
301.3	Construct various powerful parsers.	**K3**
301.4	Explain about the Intermediate code forms and type checking.	**K2**
301.5	Describe the runtime environment and Code generation.	**K2**
301.6	Discuss about code optimization techniques.	**K2**

TEXT BOOKS:

1. Compilers, Principles Techniques and Tools. Alfred V Aho, Monical S. Lam, Ravi Sethi Jeffery D. Ullman,2nd edition,pearson,2007
2. Compiler Design K.Muneeswaran, OXFORD
3. Principles of compiler design, 2nd edition, Nandhini Prasad,Elsebier.

REFERENCE BOOKS:

1. Compiler Construction, Principles and practice, Kenneth C Louden, CENGAGE
2. Implementations of Compiler, A New approach to Compilers including the algebraic methods, Yunlinsu ,SPRINGER

Course Outcome	Targeted Proficiency Level (Marks in %)	Targeted Attainment Level (Students in %)
CO1	60%	60%
CO2	60%	60%
CO3	60%	60%
CO4	60%	60%
CO5	60%	60%
CO6	60%	55%

CO1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, POs, PEOs & COs	K1	1	Lecture	BB
		Introduction to Language Processing and Lexical Analysis				
2		Describe Language processing	K1	2	Lecture	BB
3		Explain the structure of Compiler and its phases	K2	3	Lecture	BB
4		Discuss about the science of building a compiler application and Programming language basics.	K2	2	Lecture	BB
5		Explain role of Lexical analysis and buffering	K2	1	Lecture with Discussion	BB
6		Generalize specification of tokens	K2	2	Lecture	BB
7		Recognize the tokens	K2	2	Lecture	BB
8		Discuss lexical analyzer generator tool.	K2	2	Lecture	BB

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Explain the role of a Parser	K2	1	Lecture	BB
2		Discuss the Context Free Grammars	K2	1	Lecture	BB
3		Define Ambiguous, Unambiguous, Left Recursion ,Left factoring	K1	2	Lecture with Discussion	BB+ICT
4		Explain the Top Down Parser	K2	3	Lecture with Discussion	BB+ICT
5		Explain the Bottom Up Parser	K2	1	Lecture	BB+ICT
6		Describe basic structure of LR Parsers	K2	2	Lecture	BB+ICT
7		Construct LR Parsers	K3	3	Lecture with Discussion	BB+ICT

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Construct more powerful LR Parser Using Armigers Grammars	K3	3	Lecture with Discussion	BB
2		Define Syntax Directed Transaction	K1	1	Lecture with Discussion	BB
3		Discuss evolution order of SDTS	K2	2	Lecture with Discussion	BB+ICT
4		Explain Applications of SDTS	K2	1	Lecture	BB
5		Explain the SDTS	K2	1	Lecture with Discussion	BB+ICT

CO 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Explain Intermediate Code	K2	1	Lecture	BB
2		Construct 3-Address code for Variants of Syntax Trees	K2	1	Lecture with Discussion	BB+ICT
3		Discuss the Types and Declaration of Intermediate Code	K2	2	Lecture with Discussion	BB
4		Explain the Translation of Expressions	K2	1	Lecture	BB+ICT
5		Discuss Type Checking	K2	1	Lecture with Discussion	BB
6		Explain the Canted Flow Back patching	K2	1	Lecture	BB

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Define and Explain Runtime Environments	K2	1	Lecture	BB
2		Explain Stack allocation of space	K2	1	Lecture with Discussion	BB
3		Illustrate access to Non Local data on the Stack	K2	1	Lecture	BB
4		Explain Heap Management	K2	1	Lecture	BB
5		Discuss Code generation	K2	2	Lecture	BB
6		Explain Issues in design of code generation	K2	1	Lecture	BB
7		Explain the target Language Address in the target code	K2	1	Lecture	BB+ICT
8		Explain Basic blocks and Flow graphs in Code Generation	K2	2	Lecture with Discussion	BB+ICT
9		Construct Target code by Using Code generation	K2	2	Lecture with Discussion	BB+ICT

CO 6

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO6	Explain Machine Independent Optimization	K2	2	Lecture	BB
2		Discuss Principle sources of Optimization	K2	1	Lecture	BB
3		Explain Peep Hole Optimization	K2	1	Lecture with Discussion	BB
4		Define Data flow Analysis	K1	2	Lecture	BB

Total No. of Classes: 61

UNIX PROGRAMMING

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: III/I **Section: A,B,C& D**
Name of the Course: Unix programming **Course Code:R1631052**

LESSON PLAN

Course Outcomes (Along with Knowledge Level):
After completion of this course, Student will be able to:

S.NO	COURSE OUTCOMES (After completion of the course, The Learner is able to)	KNOWLEDGE LEVEL
CO1	Illustrate the Unix basics and the working of the built in commands in Unix.	K3
CO2	Illustrate the file system and change the permissions associated with files	K3
CO3	Use the concept of Shell for your own purposes	K3
CO4	Describe the Grep family and data transforming programs sed, and AWK	K2
CO5	Illustrate the working of shell programming and debugging scripts	K3
CO6	Explain the concept of process and certain essential commands	K2

TEXT BOOKS:

1. The Unix programming Environment by Brain W. Kernighan & Rob Pike, Pearson.

2. Introduction to Unix Shell Programming by M.G.Venkatesh murthy, Pearson.

REFERENCE BOOKS:

1. Unix and shell programming by B.M. Harwani, OXFORD university press.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
CO1	65	60
CO2	60	55
CO3	65	60
CO4	60	55
CO5	60	60
CO6	60	55

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO 1	Explain introduction to Unix	K1	1	Lecture	BB
2.		Describe Brief History	K1	1	Lecture	BB
3.		Enumerate What is Unix	K1	1	Lecture	BB
4.		Identify Unix Components	K2	1	Lecture	BB
5.		Explain Using Unix	K2	1	Lecture	BB
6.		Define Commands in Unix	K1	1	Lecture	BB
7.		Practice Some basic Commands	K3	1	Lecture	BB
8.		Examine Command Substitution	K3	1	Lecture	BB
9.		Practice Giving Multiple Commands	K3	1	Lecture	BB
Total :9hrs						

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO 2	Describe The Basics of Files	K1	1	Lecture	BB
2.		Describe What's in a File	K1	1	Lecture	BB
3.		Discuss Directories and File Names	K2	1	Lecture	BB
4.		Illustrate Permissions	K3	1	Lecture	BB
5.		IllustrateI Nodes	K3	1	Lecture	BB
6.		Describe The Directory Hierarchy	K1	1	Lecture	BB
7.		Practice File Attributes and Permissions	K3	1	Lecture	BB
8.		Describe File Command knowing the File Type	K1	1	Lecture	BB
9.		Practice The Chmod Command Changing File Permissions	K3	1	Lecture	BB
10.		Practice The Chown Command Changing the Owner of a File	K3	1	Lecture	BB
11.		Practice The Chgrp Command Changing the Group of a File.	K3	1	Lecture	BB
Total :11hrs						

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO3	Illustrate Command Line Structure	K3	1	Lecture	BB
2.		Demonstrate Meta characters	K3	1	Lecture	BB
3.		Practice Creating New Commands	K3	1	Lecture	BB
4.		Examine Command Arguments and Parameters	K1	1	Lecture	BB
5.		Identify Program Output as Arguments	K2	1	Lecture	BB
6.		Practice Shell Variables	K3	1	Lecture	BB
7.		Explain More on I/O Redirection	K2	1	Lecture	BB
8.		Practice Looping in Shell Programs	K3	1	Lecture	BB
Total :8hrs						

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO4	Explain The Grep Family	K2	1	Lecture	BB
2.		Explain The Stream Editor Sed	K2	1	Lecture	BB
3.		Describe The AWK Pattern Scanning and processing Language	K2	1	Lecture	BB
4.		Explain Good Files and Good Filters	K2	1	Lecture	BB
5.		Describe Other Filters	K2	1	Lecture	BB
Total :05hrs						

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO5	Practice Shell Variables	K3	1	Lecture	BB
2.		Examine The Export Command	K2	1	Lecture	BB
3.		Explain The .Profile File a Script Run During Starting	K2	1	Lecture	BB
4.		Examine The First Shell Script	K3	1	Lecture	BB
5.		Practice The read Command	K3	1	Lecture	BB
6.		Examine the Positional parameters	K3	1	Lecture	BB
7.		Explain The $? Variable knowing the exit Status	K2	1	Lecture	BB
8.		Extend More about the Set Command	K2	1	Lecture	BB
9.		Practice The Exit Command	K3	1	Lecture	BB
10.		Practice on Branching Control Structures	K3	1	Lecture	BB
11.		Practice on Loop Control Structures	K3	1	Lecture	BB
12.		Examine The Continue and Break Statement	K3	1	Lecture	BB
13.		Explain The Expr Command: Performing Integer Arithmetic	K2	1	Lecture	BB
14.		Explain Real Arithmetic in Shell programs	K2	1	Lecture	BB
15.		Explain The here Document(<<)	K2	1	Lecture	BB
16.		Practice The Sleep Command	K3	1	Lecture	BB
17.		Explain Debugging Scripts	K2	1	Lecture	BB
18.		Explain The Eval Command	K2	1	Lecture	BB
19.		Explain The Exec Command.	K2	1	Lecture	BB
Total :19hrs						

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO6	Explain parent and Child Processes	K2	1	Lecture	BB
2.		Explain Types of Processes	K2	1	Lecture	BB
3.		Extend More about Foreground and Background processes	K2	1	Lecture	BB
4.		Practice Internal and External Commands	K3	1	Lecture	BB
5.		Describe Process Creation	K2	1	Lecture	BB
6.		Explain Trap and stty Commands	K2	1	Lecture	BB
7.		- Practice The Kill Command	K3	1	Lecture	BB
8.		Explain Job Control	K2	1	Lecture	BB
Total :08hrs						

Total : 60hrs

Object Oriented Analysis and Design using UML

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: III/I **Section: A,B,C& D**
Name of the Course: OOAD Using UML **Course Code:R1631053**

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):
After completion of this course, the students will be able to:

R101053.1. Discuss Object Model. [K2]
R101053.2. Describe classes and Objects. [K2]
R101053.3. Demonstrate conceptual model of UML. [K3]
R101053.4. Develop Interaction, Use case and Activity Diagrams. [K3]
R101053.5. Illustrate advanced behavioural modeling. [K3]
R101053.6. Prepare a Case Study on any application. [K3]

TEXT BOOKS:

T1. "Object- Oriented Analysis And Design with Applications", Grady BOOCH, Robert A. Maksimchuk, Michael W. ENGLE, Bobbi J. Young, Jim Conallen, Kellia Houston, 3rd edition, 2013, PEARSON.

T2. "The Unified Modeling Language User Guide", Grady Booch, James Rumbaugh, Iva Jacobson, 12th Impression, 2012, PEARSON.

REFERENCE BOOKS:

R1. "Object-oriented analysis and design using UML", Mahesh P. Matha, PHI

R2. "Head first object-oriented analysis and design", Brett D. McLaughlin, Gary Pollice, Dave West, O"Reilly

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
R101053.1	70	70
R101053.2	65	60
R101053.3	65	60
R101053.4	60	60

R101053.5	60	60
R101053.6	60	60

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Vision, Mission of the Dept. and PEOs, Pos,& PSOs of the Programme			Lecture	BB
2		Describe the Structure of Complex systems	K1	1	Lecture	BB
3		Discuss the Inherent Complexity of Software and attributes of Complex System	K2	1	Lecture	BB
4		Describe Organized and Disorganized Complexity and bringing Order to Chaos	K2	2	Lecture With Discussion	BB
5		Describe the designing of Complex Systems	K2	2	Lecture	BB
6		Enumerate evolution and foundation of Object Model	K1	1	Lecture With Discussion	BB
7		Identify elements of Object Model	K1	1	Lecture	BB
8		Discuss the Application of Object Model	K2	1	Lecture	BB

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Describe Nature of object and Relationships among objects	K2	1	Lecture	BB
2		Describe Nature of a Class and Relationship among Classes	K2	1	Lecture with Discussion	BB
3		Relate Classes and Objects	K1	1	Lecture with Discussion	BB
4		Identify Classes and Objects	K1	1	Lecture	BB+ICT
5		Discuss Importance of Proper Classification	K2	2	Lecture with Discussion	BB+ICT
6		Explain Key abstractions and Mechanisms.	K2	2	Lecture with Discussion and in class Assignment	BB+CRE

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Explain why we model	K2	1	Lecture	BB
2		Demonstrate conceptual model of UML	K3	3	Lecture	BB
3		Describe Architecture and Classes	K2	2	Lecture with Discussion	BB
4		Discuss Relationships and Common mechanisms	K2	2	Lecture	BB+ICT
5		Develop Class and Object diagrams	K3	3	Lecture with Discussion	BB+ICT

CO 4

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Discuss interaction	K2	1	Lecture	BB
2		Develop Interaction diagrams	K3	2	Lecture with Discussion	BB
3		Discuss Use case	K2	1	Lecture with Discussion	BB
4		Develop Use case Diagrams	K3	2	Lecture with Discussion	BB+ICT
5		Develop Activity Diagrams	K3	2	Lecture with Discussion	BB+ICT

CO 5

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Discuss Events and signals	K2	2	Lecture	BB
2		Explain state machines	K2	2	Lecture with Discussion	BB
3		Discuss processes and Threads	K2	2	Lecture	BB
4		Demonstrate time and space	K3	2	Lecture with Discussion	BB
5		Develop state chart diagrams	K3	2	Lecture with Discussion	BB

CO 6

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Explain Component	K2	1	Lecture	BB
2		Explain Deployment	K2	1	Lecture with Discussion	BB
3		Develop Component diagrams	K3	2	Lecture with Discussion	BB
4		Develop Deployment diagrams	K3	2	Lecture with Discussion	BB
5		Produce Case study of Unified Library application	K3	5	Lecture with Discussion	BB

Total No. of Classes: 57

Database Management Systems

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: III/I **Section: A,B,C& D**
Name of the Course: Database Management Systems **Course Code:R1631054 (304)**

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, students will be able to:

CO	Course Outcome	Knowledge Level
C304.1	Describe database systems, various data models and database architecture	K2
C304.2	Apply ER-model to design relational databases for real time applications	K3
C304.3	Use SQL Queries, Constraints and triggers to perform database operations	K3
C304.4	Apply Normalization techniques to normalize a Database	K3
C304.4	Illustrate transaction management and concurrency control	K2
C304.5	Describe various database indexing techniques	K2

Text Books/ Reference Books suggested:

TEXT BOOKS:

- Introduction to Database Systems; C J Date; Pearson.
- Database Management Systems; Raghu Rama krishnan, Johannes Gehrke; 3/e; TMH.
- Database Systems - The Complete Book; H G Molina, J D Ullman, J Widom; Pearson.

REFERENCE BOOKS:

- Database Systems Design, Implementation and Management; Peter Rob, Carlos Coronel; 7/e.
- Database Management System; Ramez Elmasri, Shamkant B. Navathe; 6/e; PEA.
- Database System Concepts; Silberschatz, Korth; 5/e; TMH.

Targeted Proficiency Level and Targeted level of Attainment (For each course outcome):

COURSE OUTCOME	PROFICIENCY LEVEL	ATTAINMENT LEVEL
304.1	70	65
304.2	65	60
304.3	60	60
304.4	60	60
304.5	65	60
304.6	65	65

UNIT-I: AN OVERVIEW OF DATABASE MANAGEMENT & DATABASE SYSTEM ARCHITECTURE						
S.No	**Course Outcome**	**Intended Learning Outcomes (ILO)**	**Knowledge Level of ILO**	**No. of Hours**	**Pedagogy**	**Teaching aids**
1		Dissemination of vision, mission, PEOs, POs, PSOs		1	Lecture	
2	CO 1	Define a Database System, Database	K1	1	Lecture	BB
3		Describe data independence, relational systems and others	K1	2	Lecture with Discussion	BB
4		Discuss Database Architecture: The Three levels of the architecture: The External level, Conceptual level, Internal level and Mappings	K2	2	Lecture with Discussion	BB + ICT
5		Explain Database Administrator	K2	1	Lecture with Discussion	BB
6		Illustrate Database Management System Architecture and Components	K2	2	Lecture with Discussion	BB + ICT
7		Explain Client/Server Architecture	K2	1	Lecture with Discussion	BB
		TOTAL		**10**		

UNIT-II: DATABASE DESIGN, RELATIONAL MODEL, RELATIONAL ALGEBRA AND CALCULUS						
S.No	**Course Outcome**	**Intended Learning Outcomes (ILO)**	**Knowledge Level of ILO**	**No. of Hours**	**Pedagogy**	**Teaching aids**
1	CO 2	Describe introduction to Database Design: Database Design and ER Diagrams	K1	1	Lecture	BB
2		Explain Entities, Attributes and Entity Sets	K2	1	Lecture With Discussion	BB
3		Explain Relationships and Relationship Sets	K2	1	Lecture With Discussion	BB
4		Describe Relational Systems and Others	K2	2	Lecture With Discussion	BB
5		Illustrate Conceptual Design with the ER Model	K3	1	Lecture With Discussion	BB
6		Explain The Relational Model: Integrity Constraints over Relations: Key Constraints, Foreign Key Constraints, General Constraints	K2	2	Lecture With Discussion	BB
7		Illustrate Relational Algebra: Selection and Projection	K3	1	Lecture With Discussion	BB
8		Illustrate Set Operations, Renaming, Joins, Division, More Example of Algebra Queries	K3	3	Lecture With Discussion	BB
9		Illustrate Relational Calculus: Tuple Relational Calculus, Domain Relational Calculus	K3	1	Lecture With Discussion	BB
		TOTAL		**13**		

UNIT-III: QUERIES, CONSTRAINTS, TRIGGERS						
S.No	**Course Outcome**	**Intended Learning Outcomes (ILO)**	**Knowledge Level of ILO**	**No. of Hours**	**Pedagogy**	**Teaching aids**
1	CO 3	Explain the Form of a Basic SQL Query	K2	2	Lecture	BB
2		Illustrate UNION, INTERSECT and EXCEPT, Nested Queries	K3	3	Lecture With Discussion	BB
3		Illustrate Aggregate Operators	K3	2	Lecture With Discussion	BB
4		Interpret Null Values	K2	1	Lecture With Discussion	BB
5		Illustrate Complex Integrity Constraints in SQL	K3	1	Lecture With Discussion	BB
6		Illustrate Triggers and Active Databases	K3	2	Lecture With Discussion	BB
		TOTAL		**11**		

UNIT-IV: SCHEMA REFINEMENT (NORMALIZATION)						
S.No	**Course Outcome**	**Intended Learning Outcomes (ILO)**	**Knowledge Level of ILO**	**No. of Hours**	**Pedagogy**	**Teaching aids**
1	CO 4	Explain Purpose of Normalization or Schema Refinement	K2	1	Lecture With Discussion	BB + ICT
2		Explain the Concept of Functional Dependency	K2	2	Lecture With Discussion	BB + ICT
3		Illustrate Normal Forms Based on Functional Dependency (1NF, 2NF and 3 NF)	K3	3	Lecture With Discussion	BB + ICT
4		Demonstrate Concept of Surrogate Key, Boyce- Codd Normal Form (BCNF)	K3	1	Lecture With Discussion	BB + ICT
5		Illustrate Lossless Join and Dependency Preserving Decomposition, Fourth Normal Form (4NF)	K3	2	Lecture With Discussion	BB + ICT
		TOTAL		**9**		

UNIT-V: TRANSACTION MANAGEMENT AND CONCURRENCY CONTROL						
S.No	**Course Outcome**	**Intended Learning Outcomes (ILO)**	**Knowledge Level of ILO**	**No. of Hours**	**Pedagogy**	**Teaching aids**
1	CO 5	Describe Transaction, Properties of Transactions, Transaction Log	K2	1	Lecture With Discussion	BB + ICT
2		Explain Transaction Management with SQL Using Commit, Rollback and Savepoint.	K2	2	Lecture With Discussion	BB + ICT
3		Describe Concurrency Control for Lost Updates, Uncommitted Data, Inconsistent Retrievals and The Scheduler	K2	2	Lecture With Discussion	BB + ICT
4		Explain Concurrency Control with Locking Methods: Lock Granularity, Lock Types, Two Phase Locking for Ensuring Serializability	K2	3	Lecture With Discussion & Seminar	BB + ICT
5		Explain Deadlocks, Concurrency Control with Time Stamp Ordering	K2	2	Lecture With Discussion	BB + ICT
6		Explain Wait/Die and Wound/Wait Schemes	K2	1	Lecture With Discussion	BB + ICT
7		Describe Database Recovery Management: Transaction Recovery	K2	1	Lecture With Discussion	BB + ICT
		TOTAL		**12**		

UNIT-VI: OVERVIEW OF STORAGE AND INDEXING						
S.No	**Course Outcome**	**Intended Learning Outcomes (ILO)**	**Knowledge Level of ILO**	**No. of Hours**	**Pedagogy**	**Teaching aids**
1	CO 6	Describe data on External Storage	K1	1	Lecture With Discussion	BB + ICT
2		Explain File Organizations and Indexing: Clustered Indexes, Primary and Secondary Indexes	K2	2	Lecture With Discussion	BB + ICT
3		Explain Index Data Structures: Hash-Based Indexing, Tree-Based Indexing	K2	2	Lecture With Discussion	BB + ICT
4		Explain Comparison of File Organizations	K2	2	Lecture With Discussion	BB + ICT
		TOTAL		**7**		

Total No. of Classes: 62

Operating Systems

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: III/I **Section: A,B,C& D**
Name of the Course: Operating Systems **Course Code:R1631055**

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):
After completion of this course, the students will be able to:

CO 1. Describe Computer Operating System Functions, Structures and System Calls.(K1)

CO 2. Demonstrate various Process Management Concepts and CPU Scheduling Algorithms Describe Process Synchronization Techniques.(K3)

CO 3Illustrate Memory Management Techniques and Page Replacement Algorithms.(K2)

CO 4.Apply Deadlock Prevention and Avoidance Techniques.(K3)

CO 5.Demonstrate File System Concepts and Mass Storage Structures.(K3)

CO 6.Discriminate about Android platforms.(K2)

TEXT BOOK:

1. Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin and Greg Gagne 9th Edition, John Wiley and Sons Inc., 2012.
2.Operating Systems – Internals and Design Principles, William Stallings, 7th Edition, Prentice Hall, 2011.
3. Operating Systems-S Halder, Alex A Aravind Pearson Education Second Edition 2016 .

REFERENCES:

1. Modern Operating Systems, Andrew S. Tanenbaum, Second Edition, Addison Wesley, 2001.
2. Operating Systems: A Design-Oriented Approach, Charles Crowley, Tata McGraw Hill Education", 1996.
3. Operating Systems: A Concept-Based Approach, D M Dhamdhere, Second Edition, Tata
McGraw-Hill Education, 2007.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	65	65
2	65	65
3	65	65
4	60	60
5	60	60
6	60	60

Unit-I

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	**Description of Course Outcomes and Course Objectives**	**-**	**1**	Lecture	BB
1		Describe Types of operating systems	K1	1	Lecture	BB
2		Describe operating systems concepts	K1	1	Lecture	BB
3		Describe operating systems services	K1	2	Lecture	BB
4		Define Introduction to System calls	K1	1	Lecture	BB
5		Describe System call types	K1	1	Lecture	BB

Unit-II

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Demonstrate Process Concept, Process state diagram, PCB	K3	2	Lecture	BB
2		Demonstrate Process Scheduling Scheduling queues,	K3	1	Lecture with Discussion	BB
3		Demonstrate Schedulers ,Operations on Processes	K3	2	Lecture	BB
4		Demonstrate Inter process communication	K3	1	Lecture	BB
5		Demonstrate Threading Issues	K3	1	Lecture	BB
6		Scheduling Basic Concepts, Scheduling Criteria	K3	1	Lecture	BB
7		Demonstrate Scheduling Algorithms	K3	3	Lecture	BB

Unit-III

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Explain Swapping	K2	1	Lecture	BB
2		Explain Contiguous Memory Allocation	K2	1	Lecture	BB
3		Explain Paging	K2	1	Lecture	BB
4		Explain structure Of the Page Table	K2	1	Lecture	BB
5		Explain Segmentation	K2	1	Lecture	BB
6		Explain Virtual Memory	K2	1	Lecture	BB+ICT
7		Explain Demand Paging	K2	1	Lecture	BB
8		Explain Page Replacement Algorithms	K2	3	Lecture	BB+ICT
9		Explain Thrashing	K2	1	Lecture	BB

Unit-IV

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Demonstrate Process Synchronisation, Critical section Problem	K3	2	Lecture	BB
2		Demonstrate Synchronisation Hardware	K3	1	Lecture with Discussion	BB
3		Demonstrate Semaphores	K3	1	Lecture	BB
4		Demonstrate Classic problems of synchronisation	K3	1	Lecture with Discussion and in class Assignment	BB
5		Describe Monitors	K1	1	Lecture	BB
6		Demonstrate Synchronization Examples	K3	1	Lecture	BB
7		Demonstrate System model	K3	1	Lecture	BB
8		Explain Deadlock Characterization	K 2	1	Lecture	BB
9		Demonstrate Deadlock prevention ,detection and avoidance	K3	2	Lecture	BB
10		Explain Recovery from Deadlock	K2	2	Lecture	BB

Unit-V

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Define FILE and Concept of a File	K1	1	Lecture	BB
2		Demonstrate Access Methods	K3	1	Lecture with Discussion	BB
3		Demonstrate Directory Structure	K3	1	Lecture	BB
4		Explain File System mounting	K2	1	Lecture	BB
5		Explain File Sharing, Protection	K2	1	Lecture	BB
6		Demonstrate File System Structure, Allocation methods	K3	1	Lecture	BB
7		Explain Free Space Management	K2	1	Lecture	BB
8		Explain Overview of Mass Storage Structure	K2	1	Lecture	BB
9		Demonstrate Disk Scheduling, Device drivers	K3	1	Lecture	BB

Unit-VI

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Explain Components of Linux	K2	1	Lecture	BB
2		Explain Inter Process Communication	K2	1	Lecture	BB
3		Explain Synchronization and Interrupt	K2	1	Lecture	BB
4		Explain Exception and System call.	K2	1	Lecture	BB
5		Explain Android Architecture and Operating System services	K2	1	Lecture	BB+ICT
6		Explain Android Runtime Application Development	K2	2	Lecture	BB+ICT
7		Application Structure, Application Process management	K2	2	Lecture	BB+ICT

Total No. of Classes: 60

OOAD Using UML Lab

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: III/I **Section: A,B,C& D**
Name of the Course: OOAD Using UML Lab **Course Code:R1631056**

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):
After completion of this course, the students will be able to:

1	Use OOAD and UML concepts to identify Classes, Use Cases and their relationships **(K3)**
2	Develop Use case diagrams and Class Diagrams **(K3**)
3	Construct Interaction diagrams and packages**(K3)**
4	Develop State chart, Activity, Component and Deployment Diagrams **(K3)**

Operating System & Linux Programming Lab

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: III/I **Section: A,B,C& D**
Name of the Course: OS&LP Lab **Code: 307/R1631057**

LESSON PLAN

Course Outcomes

Upon Completion of the course, student will be able to:

S.NO	COURSE OUTCOMES (After completion of the course, The Learner is able to)	KNOWLEDGE LEVEL
307.1	Illustrate CPU scheduling algorithms.	K3
307.2	Apply Bankers Algorithm for Dead Lock Avoidance and Dead Lock Prevention.	K3
307.3	Use Page replacement algorithms for memory management.	K3
307.4	Apply Unix/Linux concepts and commands.	K3
307.5	Use shared memory for inter-process communication.	K3
307.6	Use pthreads library for multithreading.	K3

Database Management Systems Lab

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: III/I **Section: A,B,C& D**
Name of the Course: Database Management Systems Lab **Course Code: R1631058/308**

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, students will be able to:

CO	COURSE OUTCOME	KNOWLEDGE LEVEL
308.1	Understand and apply common SQL statements and perform different operations	K3
308.2	Design data model from specifications given and also perform the action of transforming the data models to other.	K3
308.3	Understand and apply the normalization techniques	K3
308.4	Understand and able to access and manipulate data using PL/SQL blocks.	K3

Professional Ethics and Human values

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: III/I **Section: A,B,C& D**
Name of the Course: Professional Ethics and Human values
Course Code:R1631049/C309

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1. Recollect the human, moral values and ethics.[K1]

CO2. Illustrate the principles to being harmony among I, we and nature by focusing on human duties, rights, and dignity.[K2]

CO3. Describe the various Engineering Ethics and social issues that are encountered by every professional in discharging professional duties.[K2]

CO4. Describe the Engineers' Responsibilities towards Safety and Risk and based on this make analysis on designing to keep safety measure**.** [K2]

CO5. Demonstrate the professional ethics and techniques for collegiality and problem solving [K3]

CO6. Discuss the globalization and MNC issues like cross culture , business ethics and Research ethics. [K2]

TEXT BOOKS:

T1. Engineering Ethics and Human Values by M.Govindarajan, S.Natarajan and V.S.Senthil Kumar- PHI Learning

T2. Professional Ethics and Morals by Prof. A.R.Aryasri , Dharani kota Suyodhana- Maruthi Publications.

T3. Professional Ethics and Human Values by A.Alavudeen, R.Kalil Rahmanand M.Jaya kumaran- Laxmi Publications.

REFERENCE BOOKS:

R1. Professional Ethics by R. Subramaniam – Oxford Publications, New Delhi.

R2. Ethics in Engineering by Mike W. Martin and Roland Schinzinger - Tata McGraw-Hill

R3. Professional Ethics and Human Values by Prof. D.R. Kiran-Tata McGraw-Hill Publications.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	60	70
2	60	70
3	60	60
4	60	70
5	60	60
6	60	60

Unit 1

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Vision, Mission of the Dept. and PEOs, POs,& PSOs of the Programme			Lecture	BB
2		Define the Moral values and Ethics, Integrity and Trustworthiness, Work Ethics, Service Learning	K1	1	Lecture	BB
3		Identify Civic Virtue, Respect for others	K1	1	Lecture	BB
4		Define how living peacefully, caringly, Sharing and become Honesty, Courage	K1	1	Lecture	BB
5		Define how to value Time, Co-operation, Commitment, become Empathy, Self-confident, Spirituality, Character	K1	1	Lecture	BB

Total 4

Unit 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Classify the Truth fulness, Customs and Traditions.	K2	1	Lecture	BB
2		Describe the Value Education, Human Dignity.	K2	1	Lecture	BB
3		Explain Human Rights, Fundamental Duties, Aspirations and Harmony (I, We & Nature).	K2	1	Lecture	BB
4		Clarify the Gender Bias, Emotional intelligence.	K1	1	Lecture	BB
5		Explain Salovey, Mayer Model.	K2	1	Lecture	BB
6		Describe Emotional Competencies and Conscientiousness.	K2	1	Lecture	BB

Total 6

Unit 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Explain the History of Ethics, Need of Engineering Ethics.	K2	1	Lecture	BB
2		Distinguish Senses of Engineering Ethics, Profession and Professionalism	K2	1	Lecture	BB
3		Discuss Self Interest, Moral Autonomy , Utilitarianism, Virtue Theory	K2	1	Lecture with Discussion	BB
4		Explain the Uses of Ethical Theories, Deontology, Types of Inquiry, Kohlberg's Theory	K2	1	Lecture	BB
5		Explain Gilligan's Argument, Heinz's Dilemma	K2	1	Lecture	BB
6		Comparison with Standard Experiments, Learning from	K2	1	Lecture	BB

		the Past ,Engineers as Managers				
7		Describe Consultants and Leaders, Balanced Outlook on Law, Role of Codes, Codes and Experimental Nature of Engineering.	K2	1	Lecture	BB

Total **7**

Unit 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Explain the Concept of Safety, Safety and Risk, Types of Risks, and Voluntary v/s Involuntary Risk Consequences	K2	1	Lecture	BB
2		Discuss the Risk Assessment Accountability, Liability	K2	1	Lecture with Discussion	BB
3		Describe Levels of Risk, Delayed v/s Immediate Risk	K2	1	Lecture	BB
4		Explain the Safety and the Engineer , Designing for Safety	K2	1	Lecture	BB
5		Illustrate the risk Benefit Analysis, Accidents	K2	1	Lecture	BB

Total **5**

Unit 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Explain the Concept of Duty and Professional Duties	K2	1	Lecture	BB
2		Demonstrate the Collegiality, Techniques for Achieving Collegiality	K3	1	Lecture	BB
3		Explain the Senses of Loyalty,	K2	1	Lecture	BB

		Consensus and Controversy				
4		Identify the Professional and Individual Rights	K1	1	Lecture	BB
5		Illustrate the Confidential and Proprietary Information, Conflict of Interest-Ethical egoism, Collective Bargaining, Confidentiality	K2	1	Lecture	BB
6		List the Gifts and Bribes and Problem solving, Occupational Crimes, Industrial Espionage-Price Fixing-Whistle Blowing	K1	1	Lecture	BB

Total 6

Unit 6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Describe the Globalization and MNCs, Cross Culture Issues , Business Ethics	K2	1	Lecture	BB
2		Discuss the Endangering Lives, Bio Ethics	K2	1	Lecture with discussion	BB
3		Explain Computer Ethics, War Ethics	K2	1	Lecture	BB
4		Define the Research Ethics, Intellectual Property Rights. Media Ethics, Environmental Ethics	K1	1	Lecture	BB

Total 4

Total No. of Classes: 32